

Total marks – 70**QUESTION 1-10 Objective-response questions.****(10marks)****Write answers on the Objective response answers booklet****Multiple Choice**

1. For what value of k is $x - 1$ a factor of $2x^3 - 3x^2 + kx - 5$

(A) -10 (B) -4
(C) 6 (D) 10

2. The point P divides the interval from A(-2,2) to B(8,-3) internally in the ratio 3:2
What is the x-coordinate of P?

(A) 4 (B) 2
(C) 0 (D) -1

3. What is the value $\sin 2\theta$, given $\sin \theta = \frac{3}{5}$ and $\sin \theta > 0$?

(A) $\frac{6}{5}$ (B) $\frac{24}{25}$ (C) $\frac{4}{5}$ (D) $\frac{12}{25}$

4. When the polynomial $P(x)$ is divided by $x^2 - 2x - 3$ the remainder is $2x - 5$.
What is the remainder when $P(x)$ is divided by $x + 1$?

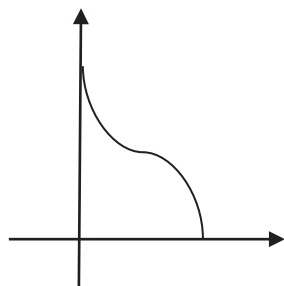
(A) -11 (B) -7 (C) -3 (D) 1

5. $\tan(\alpha - \beta) =$

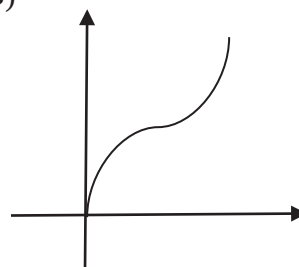
(A) $\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$ (B) $\frac{\tan \alpha \times \tan \beta}{1 - \tan \alpha \tan \beta}$
(C) $\frac{\tan \beta - \tan \alpha}{1 + \tan \beta \tan \alpha}$ (D) $\frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

6. Which graph best represents $y = \cos^{-1}(1 - x)$

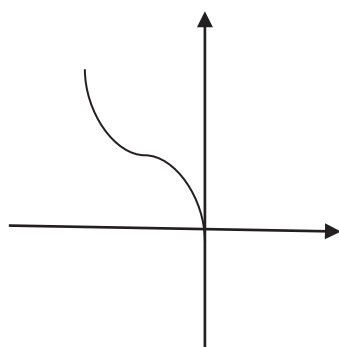
(A)



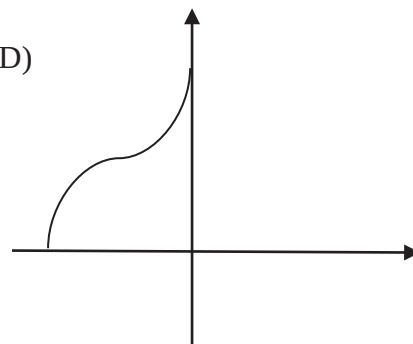
(B)



(C)



(D)



7. $\int_0^{\pi} \frac{\cos x}{\sqrt{1 - \sin^2 x}} dx$ is not equal to

(A) $\left[\sin^{-1}(\cos x) \right]_0^{\pi}$

(B) $\int_0^{\pi} 1 dx$

(C) $-\pi$

(D) $\left[\ln(\cos x) \right]_0^{\pi}$

8. Which is not a polynomial

(A) $x^3 - 5x + 1$

(B) $x^4 + 3x^3 - \sqrt{3}x^2 + x - 1$

(C) $x^3 + 3x^2 + 2x - 3x^{-1}$

(D) $(3x - 1)^3$

9. The general solution to $\sqrt{3} \sin x - \cos x = 0$ where n is an integer is?

(A) $\frac{\pi}{3} + n\pi$

(B) $\frac{(6n+1)\pi}{6}$

(C) $\frac{\pi}{3} + 2n\pi$ or $\frac{2\pi}{3} + 2n\pi$

(D) $\frac{\pi}{6} + 2n\pi$ or $\frac{2\pi}{6} + 2n\pi$

10. Which expression best represents the primitive function for $y = \cos^2 x$.

(A) $\frac{1}{4} \sin 2x + \frac{x}{2} + c$

(B) $\frac{1}{2} (\sin 2x + x) + c$

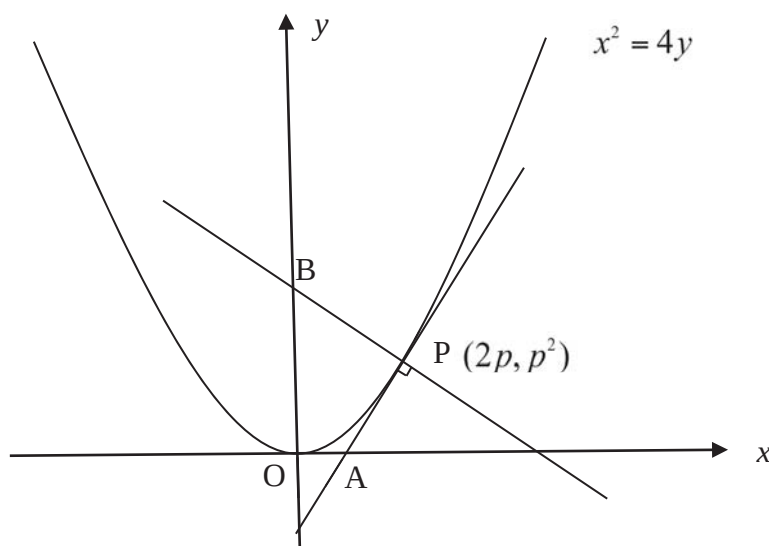
(C) $\frac{1}{4} \sin^2 x + c$

(D) $\frac{1}{4} \sin 2x - \frac{x}{2} + c$

End of Question Multiple Choice

QUESTION 11 (16 marks) **START A NEW BOOKLET** **Mark**

- (a) For $A(3, -1)$ and $B(a, 2)$, the point $(13, 4)$ divides the interval AB externally in the ratio $5:2$. Find the value of a . 2
- (b) A curve has parametric equations $x = \frac{t}{3}, y = 2t^2$. Write down the Cartesian equation for this curve. 2
- (c) Solve the inequality $\frac{3}{x-2} \leq 1$ 3
- (d) For $n = 1, 2, 3, \dots$ let $S_n = 1^2 + 2^2 + \dots + n^2$. 3
 Use mathematical induction to prove $S_n = \frac{1}{6}n(n+1)(2n+1)$.
- (e) The diagram shows the graph of the parabola $x^2 = 4y$. The tangent to the parabola at $P(2p, p^2)$, $p > 0$, cuts the x axis at A . The normal to the parabola at P cuts the y axis at B



- (i) Derive the equation of the tangent AP . 2
- (ii) Show that B has coordinates $(0, p^2 + 2)$. 2
- (iii) Let C be the midpoint of AB . Find the Cartesian equation of the locus of C . 2

QUESTION 12 (13 marks) **START A NEW BOOKLET****Marks**

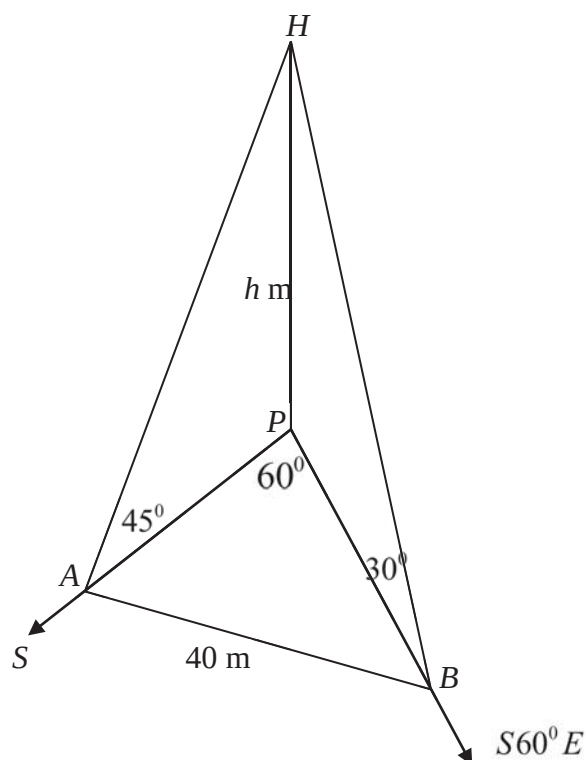
(a) (i) Express $\sin x - \cos x$ in the form $R\sin(x - \alpha)$. 2

(ii) Hence solve the equation $\sin x - \cos x = \frac{\sqrt{6}}{2}$, $0 \leq x \leq 2\pi$. 2

(b) Evaluate $\int_0^{\frac{\pi}{4}} 2\sin^2 x dx$. 2

(c) Using the substitution $t = \tan \frac{\theta}{2}$, show $\frac{\sin \theta}{1 - \cos \theta} = \cot \frac{\theta}{2}$. 2

(d)



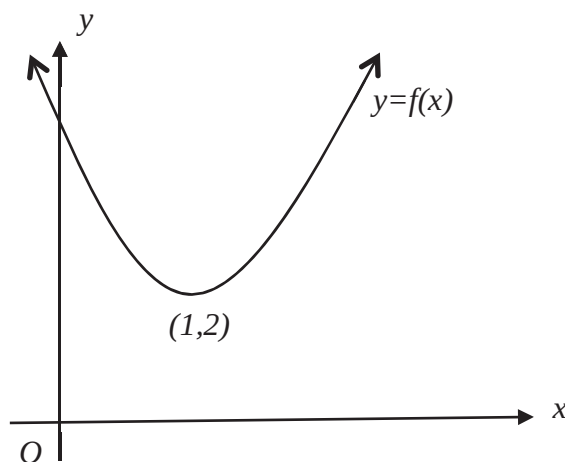
From point A the angle of elevation to the top of the tower HP is 45° . A is due south of P. From point B the angle of elevation to the top of the tower is 30° and B is $S60^\circ E$ of P. AB is a distance of 40 metres.

(i) Show that $BP = \sqrt{3}h$ and $AP = h$, where the tower is h metres high. 2

(ii) Show that $h = \sqrt{\frac{1600}{4 - \sqrt{3}}}$ metres. 3

QUESTION 13 (15 marks) **START A NEW BOOKLET****Marks**

- (a) The graph of $f(x) = x^2 - 2x + 3$ is shown in the diagram.



- (i) Explain why $f(x)$ does not have an inverse function. 1
- (ii) Sketch the graph of the inverse function $g^{-1}(x)$ of $g(x)$, where $g(x) = x^2 - 2x + 3, x \geq 1$. 1
- (b) (i) Find the domain and range for the function $y = 3\sin^{-1} \frac{x}{2}$. 2
- (ii) Sketch $y = 3\sin^{-1} \frac{x}{2}$. 2
- (c) Find the exact values of each of the following, showing all working.
- (i) $\sin^{-1} \left(\sin \frac{4\pi}{3} \right)$ 1
- (ii) $\sin \left(2 \cos^{-1} \frac{1}{3} \right)$. 2
- (d) (i) Differentiate $\sin^{-1} \frac{1}{2} x^3$. 2
- (ii) Hence find $\int \frac{x^2}{\sqrt{4-x^6}} dx$ 2
- (e) Evaluate $\int_0^3 \frac{5}{9+x^2} dx$ 2

QUESTION 14 (16 marks) **START A NEW BOOKLET** **Marks**

- (a) When $P(x) = 3x^3 - x^2 + x + a$ is divided by $(x - 1)$ the remainder is 3. **1**
Find a .

- (b) The polynomial $P(x) = x^3 + bx^2 + cx + d$ has roots $-1, 2, 3$. Find b, c , and d . **3**

- (c) Sketch the following polynomial. Clearly show all intercepts.

$$P(x) = x(x - 3)^2(x + 2)^3. \quad \mathbf{3}$$

- (d) The polynomial equation $x^3 + 7x^2 - 2x + 3 = 0$ has 3 roots, α, β, γ

(i) Find $\alpha + \beta + \gamma$ **1**

(ii) Find $\alpha\beta + \beta\gamma + \gamma\alpha$ **1**

(iii) Find $\alpha^2 + \beta^2 + \gamma^2$ **2**

- (e) It is known that two of the roots of the equation $2x^3 + x^2 - kx + 6 = 0$ are reciprocals of each other. Find the value of k . **2**

- (f) Given that $(x - 1)$ and $(x + 3)$ are factors of $P(x) = x^4 - 2x^3 - 32x^2 - 30x + 63$, **3**
express $P(x)$ in factorised form.

END OF THE PAPER

Q11

(a) use ratios $5:-2$.

$$\left(\frac{-2 \times 3 + 5 \times a}{5-2}, \right) = (13, 4)$$

$$\therefore \frac{-6 + 5a}{3} = 13.$$

$$-6 + 5a = 39$$

$$5a = 45$$

$$a = 9 \quad \checkmark$$

many did not know formula.

$$(b) x = \frac{t}{3} \quad y = 2t^2.$$

$$t = 3x \quad \checkmark \therefore y = 2(3x)^2$$

$$y = 18x^2 \quad \checkmark$$

Done well.

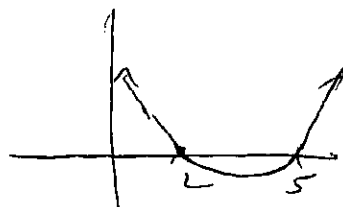
$$(c) x(x-2)^2 \quad 3(x-2) \leq (x-2)^2 \quad \checkmark$$

$$x \neq 2 \quad 3(x-2)^2 - 3(x-2) \geq 0$$

$$(x-2)(x-2-3) \geq 0 \quad \checkmark$$

$$(x-2)(x-5) \geq 0 \quad \checkmark$$

$$x < 2 \quad x \geq 5 \quad \checkmark$$

(many forgot $x \neq 2$)

$$(d) 1^2 + 2^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1)$$

Show true for $n=1$

$$\text{LHS} = 1^2$$

$$\text{RHS} = \frac{1}{6} \times 1 \times 2 \times 3$$

$$= 1$$

$$\text{LHS} = \text{RHS} \therefore \text{true for } n=1$$

Let it be true for $n=k$

$$\therefore 1^2 + 2^2 + \dots + k^2 = \frac{1}{6} k(k+1)(2k+1) \quad *$$

show true for $n=k+1$

$$1^2 + 2^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6} (k+1)(k+2)(2k+3)$$

$$\text{LHS} = 1^2 + 2^2 + \dots + k^2 + (k+1)^2$$

$$= \frac{1}{6} (k(k+1)(2k+1) + 6(k+1)^2)$$

$$= \frac{1}{6} (k+1) [k(2k+1) + 6(k+1)]$$

$$= \frac{1}{6} (k+1) (2k^2 + k + 6k + 6)$$

$$= \frac{1}{6} (k+1) (2k^2 + 7k + 6)$$

$$= \frac{1}{6} (k+1) (k+2)(2k+3)$$

$$= \text{RHS}$$

Many expanded at this point

On the whole poorly done for such an easy induction question.

 $\therefore$ By mathematical induction true for all $n \geq 1$

$$(c) (i) y = \frac{x^2}{4}$$

$$\frac{dy}{dx} = \frac{x}{2}$$

This whole question was generally done well.

At $x=2p$

$$\frac{dy}{dx} = p. \checkmark$$

- some quoted this result so did not get this mark.

Equation of tangent

$$y - p^2 = p(x - 2p)$$

$$y - p^2 = px - 2p^2$$

$$y = px - p^2 \checkmark$$

(ii) ~~also~~ gradient of normal is $-\frac{1}{p}$.

Equation of normal $y - p^2 = -\frac{1}{p}(x - 2p) \checkmark$

When $x=0$ $y - p^2 = -\frac{1}{p}x - 2p \checkmark$

$$y = p^2 + 2$$

(could be shown by using $y = mx + b$)

$\therefore B$ is $(0, p^2 + 2)$

(iii) A is where $y=0$ on tangent.

i.e. $px - p^2 = 0$

$$px = p^2$$

$$x = p$$

i.e. A is $(p, 0)$.

mid point of AB is $(\frac{p}{2}, \frac{p^2+2}{2}) \checkmark$

i.e. $x = \frac{p}{2}$ $y = \frac{p^2+2}{2}$

$p = 2x$ $y = \frac{(2x)^2+2}{2}$

$$y = \frac{4x^2+2}{2}$$

$$y = 2x^2 + 1. \checkmark$$

Q12

a) i)

$$R \sin(x-a) = R (\sin x \cos a - \cos x \sin a)$$

$$\begin{aligned} \sin x - \cos x &= \sqrt{2} \left(\sin x \frac{1}{\sqrt{2}} - \cos x \frac{1}{\sqrt{2}} \right) \\ &= \sqrt{2} \sin \left(x - \frac{\pi}{4} \right) \checkmark \end{aligned}$$

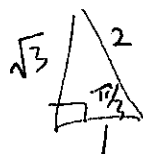
ii)

$$\sqrt{2} \sin \left(x - \frac{\pi}{4} \right) = \frac{\sqrt{6}}{2}$$

$$\sin \left(x - \frac{\pi}{4} \right) = \frac{\sqrt{3}}{2}$$

$$x - \frac{\pi}{4} = \frac{\pi}{3}, \frac{2\pi}{3} \checkmark$$

$$x = \frac{7\pi}{12}, \frac{11\pi}{12} \checkmark$$



b)

$$\begin{aligned} \int_0^{\pi/4} 2 \left(\frac{1}{2} (1 - \cos 2x) \right) dx &= \int_0^{\pi/4} \left(\frac{1}{2} - \frac{1}{2} \cos 2x \right) dx \\ &= \left[x - \frac{1}{2} \sin 2x \right]_0^{\pi/4} \checkmark \\ &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\ &= \frac{\pi}{4} - \frac{1}{2} \checkmark \end{aligned}$$

c)

$$\sin \theta = \frac{2t}{1+t^2}, \quad \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\begin{aligned} \text{LHS} &= \frac{\sin \theta}{1 - \cos \theta} \\ &= \frac{\frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}} \checkmark \end{aligned}$$

$$= \frac{2t}{1+t^2 - (1-t^2)} \checkmark$$

$$= \frac{2t}{2t^2}$$

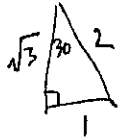
$$= \frac{1}{t}$$

$$= \cot \frac{\theta}{2}$$

$$= \text{RHS}$$

$$d) i) \tan 30^\circ = \frac{h}{BP} \checkmark$$

$$BP = \frac{h}{\tan 30^\circ} \\ = \sqrt{3}h$$



$$ii) \tan 45^\circ = \frac{h}{AP} \checkmark \\ AP = \frac{h}{\tan 45^\circ} \\ = h$$

ii)

$$AB^2 = AP^2 + BP^2 - 2 \times AP \times BP \cos 60^\circ \checkmark$$

$$40^2 = h^2 + (\sqrt{3}h)^2 - 2 \times h \times \sqrt{3}h \times \frac{1}{2} \checkmark$$

$$1600 = h^2 + 3h^2 - \sqrt{3}h^2$$

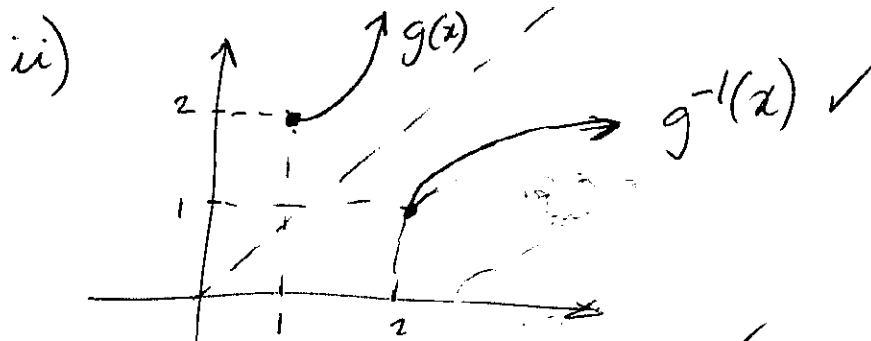
$$1600 = 4h^2 - \sqrt{3}h^2$$

$$h^2(4 - \sqrt{3}) = 1600 \checkmark$$

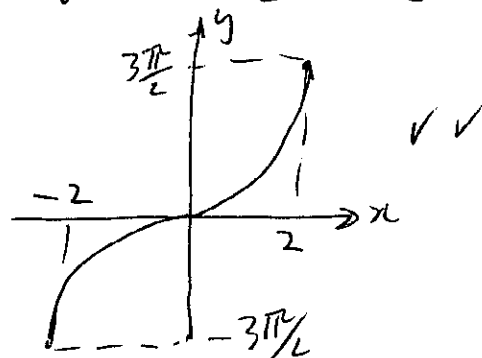
$$h^2 = \frac{1600}{4 - \sqrt{3}}$$

$$h = \sqrt{\frac{1600}{4 - \sqrt{3}}}$$

Q13 a) i) - there are y values which have two x values ✓
 - Horizontal line test

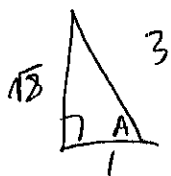


b) range $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$ domain $-2 \leq x \leq 2$ ✓



c) i) $\sin^{-1}(\sin \frac{4\pi}{3}) = -\frac{\pi}{3}$ ✓

ii) $\sin(2 \cos^{-1} \frac{1}{3}) = 2 \sin A \cos A$
 $= 2 \times \frac{\sqrt{8}}{3} \times \frac{1}{3}$
 $= \frac{4\sqrt{2}}{9}$



d) $\frac{d}{dx} \sin^{-1} \frac{1}{2} x^3 = \frac{1}{\sqrt{1-(\frac{x^3}{2})^2}} \times \frac{3x^2}{2}$ ✓
 $= \frac{3x^2}{2\sqrt{1-\frac{x^6}{4}}} = \frac{3x^2}{\sqrt{4-x^6}}$ ✓

Q13 d) iv $\int \frac{x^2}{\sqrt{4-x^6}} dx = \frac{1}{3} \int \frac{3x^2}{\sqrt{4-x^6}} dx \checkmark$
 $= \frac{1}{3} \sin^{-1} \frac{1}{2} x^3 + C \checkmark$

e) $\int_0^3 \frac{5}{9+x^2} dx = \frac{5}{3} \left[\tan^{-1} \frac{x}{3} \right]_0^3 \checkmark$
 $= \frac{5}{3} (\tan^{-1} 1 - \tan^{-1} 0)$
 $= \frac{5\pi}{12} \checkmark$

(a) $a = 0$

(b) $-1 + 2 + 3 = -b$

$\therefore b = -4$

$-1 \cdot 2 \cdot 3 = -d$

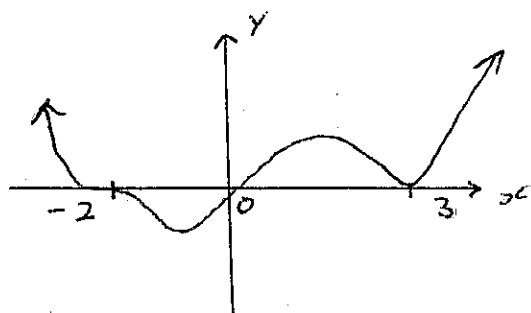
$\therefore d = 6$

$(-1)^3 - 4(-1)^2 + c(-1) + 6 = 0$

$-1 - 4 - c + 6 = 0$

$\therefore c = 1$

(c)



(d) (i) -7

(ii) -2

(iii) $(-7)^2 - 2(-2) = 53$

(e) Let roots be $\alpha, \frac{1}{\alpha}, \beta$

$\alpha \cdot \frac{1}{\alpha} \cdot \beta = -\frac{6}{2}$ — (1)

$\therefore \beta = -3$

$\alpha + \frac{1}{\alpha} + \beta = -\frac{1}{2}$ — (2)

$\therefore \alpha + \frac{1}{\alpha} = \frac{5}{2}$

$\alpha \cdot \frac{1}{\alpha} + \alpha \cdot \beta + \frac{1}{\alpha} \cdot \beta = -\frac{k}{2}$

$\therefore k = -2(1 + \beta(\alpha + \frac{1}{\alpha}))$

$= -2(1 - 3(\frac{5}{2}))$

$= 13$

(f) $1, -3, \alpha, \beta = 63 \Rightarrow \alpha\beta = -21$

$1 - 3 + \alpha + \beta = 2 \Rightarrow \alpha + \beta = 4$

by inspection, $\alpha = 7, \beta = -3$

Marks

(a) 1 mark for answer

(b) 1 mark for each answer.

(c) 1 mark for a "correct" intercept at $x = -2$ or $x = 3$.

2 marks for two "correct" intercepts at $x = 2$ and $x = 3$.

3 marks for correct graph.

(d) (i) 1 mark for answer

(ii) 1 mark for answer

(iii) 1 mark for demonstrating the use of:

$x^2 + y^2 + z^2 = (x + y + z)^2 - 2(xy + xz + yz)$

1 mark for "correct" answer.

(e) 1 mark for $\beta = -3$
1 mark for $k = 13$

(f) Most (all?) students used long division.

1 mark for making reasonable progress with long division.

2 marks for correct division of $P(x)$ by $(x-1)(x+3)$

3 marks for correct answer.

$\therefore$
 $P(x) = (x-1)(x+3)(x-7)(x+3)$
 $= (x-1)(x-7)(x+3)^2$